

SUMMARY

AI & Machine Learning Engineering student (CGPA: 8.97/10) with experience building and deploying end-to-end machine learning applications. Skilled in Python, FastAPI, PyTorch, TensorFlow, and Docker, with project experience spanning NLP, MLOps, and predictive analytics. Passionate about turning data-driven ideas into practical solutions and eager to contribute to real-world AI products while continuing to grow as an engineer.

SKILLS

Languages: Python, Java, JavaScript, SQL, C

AI / Machine Learning: NLP, Text-to-SQL, LSTM, Scikit-learn, PyTorch, TensorFlow

Frameworks & Libraries: FastAPI, React, Next.js, SQLAlchemy, SciPy, PyAutoGUI

Databases & DevOps: PostgreSQL, Supabase, Docker, Docker Compose, Cloudflare

Developer Tools: Git & GitHub, VS Code, ESP32-S3

Soft Skills: Communication, Problem Solving, Team Collaboration, Research and Documentation

WORK EXPERIENCE

VedNova AI Technologies LLP | Software Engineer Intern

July 2026–Present | Remote

Next.js, React, TypeScript, Tailwind CSS, Supabase, PostgreSQL, Git, Cloudflare

- Diagnosed and resolved **critical frontend-backend integration defects** in a production Next.js, React, TypeScript, and Supabase application, directly improving **application stability and end-user experience**.
- Shipped features end-to-end across both the customer-facing website and internal admin dashboard, resolving issues spanning **REST API integration, PostgreSQL database queries, authentication flows, and UI rendering**.
- Performed full-stack debugging and validated **CRUD workflows and backend connectivity**, delivering **production-ready fixes** while collaborating through Git-based version control in an agile team environment.

PROJECTS

AI-Powered Air Quality Monitoring System (ISRO Project)

Nov 2025 – Present

Python, LSTM, SciPy, C++, ESP32-S3

- Collaborating as a **System Engineer and Data Analyst Project Intern** on a next-generation edge-computing Air Quality Monitoring System (AQMS) for the **Indian Space Research Organisation (ISRO)**, coordinating across hardware, physics-modeling, and ML sub-teams.
- Co-developing a hybrid temporal forecasting pipeline powered by an **LSTM network** that ingests physics-validated atmospheric outputs, enabling **real-time air quality prediction and automated regulatory compliance tracking**.
- Assisting in migrating the system's core simulation engine from a legacy **Physics-Informed Neural Network (PINN)** to a deterministic **Chemical Transport Model (CTM)**, applying **operator-splitting techniques** to improve simulation accuracy and reduce computational overhead.
- Integrating an **ESP32-S3 microcontroller** with I2C/UART expansion bridges to build a **zero-loss, non-blocking sensor data ingestion pipeline**, ensuring uninterrupted real-time data feeds to the downstream ML model.

ModelOps Studio | [Live Demo](#)

June 2026

Python, FastAPI, Scikit-learn, SQLAlchemy, Docker, Plotly, Jinja2, SQLite

- Architected and built a full-stack, Vertex AI-inspired MLOps platform end-to-end, spanning dataset ingestion, exploratory analysis, model training, comparison, deployment, and monitoring, exposing **16 REST API endpoints** through a FastAPI backend and server-rendered UI, with average API response latency of **~220ms**.
- Automated training and evaluation pipelines across **5 ML algorithms** with a shared preprocessing layer and 3-fold cross-validation, powering **80+ training runs** on datasets of up to **30,000 records** while dynamically detecting classification vs. regression tasks and versioning every run with metrics, hyperparameters, and logs.
- Designed and implemented a model-monitoring service using Kolmogorov-Smirnov and Population Stability Index statistical tests, achieving **97% accuracy** in detecting data drift in production and automatically surfacing retraining recommendations.
- Containerized the entire application with Docker and deployed it to production on Render, cutting end-to-end ML workflow time by **~60-70%** versus a manual process, backed by a database-agnostic SQLAlchemy ORM layer portable across SQLite and PostgreSQL.

NLP Query Engine | [Live Demo](#)

Jan 2026

Python, FastAPI, React, PostgreSQL, Docker

- Architected a full-stack natural-language-to-SQL engine exposing **12 REST API endpoints** across three entry points, direct PostgreSQL connections, file uploads, and an offline demo dataset, unified under a single pluggable data-source abstraction layer.
- Built a dynamic schema-discovery system tested against **20 database tables and 55,000+ rows**, automatically introspecting any connected SQL backend at runtime, including CSV and XLSX files loaded into an in-memory SQLite database, with no hard-coded schema definitions.
- Designed a hybrid query classifier achieving **88% natural-language-to-SQL accuracy** and a **92% query execution success rate**, routing each question to LLM-based Text-to-SQL generation, retrieval-augmented search over **150 indexed documents (3,200+ chunks)**, or both.
- Engineered a single-container production deployment supporting **7+ data source formats** with a **1.1s average response time**, serving a compiled React frontend and FastAPI backend from one port, containerized with Docker and deployed live on Render.

CERTIFICATIONS

Artificial Intelligence: Knowledge Representation and Reasoning — **NPTEL**

Gen AI Powered Data Analytics Job Simulation — **TATA iQ, Forage**

AI Fluency: Framework and Foundations — **Anthropic**

EDUCATION

Cambridge Institute of Technology

2023 – 2027

Bachelor of Engineering (AI & ML) CGPA: 8.97/10